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Property Testing using Python's Hypothesis

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Schedule

9:30	Introduction. Testing simple functions
10:15	*Coffee break*
10:35	Practical 1
11:35	Testing functions with structured data
12:30	*Lunch break*
14:00	Practical 2
15:00	Testing stateful code
15:30	*Coffee break*
15:50	Practical 3
16:30	End of the workshop

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Overview

- A long-standing challenge for software engineering is ensuring software correctness
- Formal verification is (still) expensive and rarely used
- *Tests* are the most commonly used practical technique
- *Unit tests* are the industry-standard for checking software components

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This workshop

- Introduction to *property-based testing*
- An automatic testing methodology that complements unit tests
- We will use Python (but it is available for many other programming languages)
- Many successful applications in both open-source and some industrial projects
- But still not commonly known and under-utilized in practice

Slides and demo code: `https://github.com/pbv/pbt-workshop`

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Intended audience

Familiarity with basic Python programming:

- built-in data types and structures (lists, tuples, dictionaries)
- defining functions and methods
- basics of functional programming (map, filter, lambdas)

Familiarity with basics of testing:

- assertions
- defining unit tests for simple functions
- running tests automatically (pytest, VSCode)

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Unit tests

- Code fragments for testing functions, methods, classes, libraries, etc.
- Express the expected behaviour for specific combinations of inputs

Unit tests (cont.)

Example: testing an integer square root function in Python

```
import math

def isqrt(n):
    return int(math.sqrt(n))

def test_isqrt():
    assert isqrt(0) == 0
    assert isqrt(2) == 1
    assert isqrt(4) == 2
    assert isqrt(5) == 2
    assert isqrt(9) == 3
```

Problems with unit tests

Cognitive bias:

- how can we include an edge case in the tests that we didn't consider in the code?

Poor scaling:

- a few unit tests per feature
- for n features, $O(n)$ unit tests
- but testing *interactions* between features requires $O(n^2), O(n^3), \ldots$ unit tests

Problems with unit tests

Cognitive bias:

- how can we include an edge case in the tests that we didn't consider in the code?

Poor scaling:

- a few unit tests per feature
- for n features, $O(n)$ unit tests
- but testing *interactions* between features requires $O(n^2), O(n^3), \ldots$ unit tests

Solution:

> *"Don't write tests — generate them!"*

John Hughes, co-author of the *QuickCheck* library

Property-based testing

- Write properties instead of specific tests
  - functions that take some arguments and assert something
  - should be universal, i.e. hold for *all* values
  - should define the expected behaviour for *all* cases
- Specify *generators* for the inputs
- The testing framework executes the property with a large number of inputs
  - testing fails if a counter-example is found
  - if no counter-example is found, testing succeeds

Property-based testing (cont.)

- QuickCheck was the first PBT library (for Haskell)
- Hence, PBT is sometimes called "QuickCheck testing"
- Other PBT libraries:

PropEr	for Erlang
ScalaCheck	for Scala
FsCheck	for F#
JUnit-QuickCheck	for Java
RapidCheck	for C++
Hypothesis	for Python

Many others: `https://en.wikipedia.org/wiki/QuickCheck`

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Examples of industrial adoption

- Ericsson used PBT to test telecoms software and found hidden bugs in production code
- Volvo used PBT to test automotive software find bugs in both code and specifications
- Dropbox used PBT to test distributed file synchronization
- Many Python open-source projects are tested using Hypothesis e.g. pytorch, jax, numpy, pandas, mercurial

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From unit tests to properties

Let us look back to our unit tests for the integer square root:

```
def test_isqrt():
    assert isqrt(0) == 0
    assert isqrt(2) == 1
    assert isqrt(4) == 2
    assert isqrt(5) == 2
    assert isqrt(9) == 3
```

How can we change this into a property?

A false start

```
def test_isqrt(n):  # for some unknown n
    assert isqrt(n) == ????
```

- How can we fill in the blanks?

A false start

```
def test_isqrt(n):  # for some unknown n
    assert isqrt(n) == guess_isqrt(n)
```

- How can we fill in the blanks?
- We need to "guess" the result of `isqrt`

A false start

```
def test_isqrt(n):  # for some unknown n
    assert isqrt(n) == guess_isqrt(n)
```

- How can we fill in the blanks?
- We need to "guess" the result of `isqrt`
- But that is just as much work as the function we are trying to test!

A property about square roots

Let n be some non-negative number and $r = \texttt{isqrt}(n)$; then r should be largest non-negative integer such that $r^2 \leq n$.

Mathematically:

$$r \geq 0 \wedge r^2 \leq n \wedge (r+1)^2 > n$$

A property about square roots (cont.)

Using Python and Hypothesis:

```
from hypothesis import given
import hypothesis.strategies as st

@given(st.integers(min_value=0))
def test_isqrt(n):
    r = isqrt(n)
    assert r>=0 and r**2<=n and (r+1)**2>n
```

(Cue demo.)

A property about square roots (cont.)

- The property failed for a very large number
- This happens because `math.sqrt` converts integers to floating point numbers, thereby losing precision
- We hadn't anticipated this edge case in our unit tests
- How to fix this?

  change the property restricting generation to smaller numbers

```
@given(st.integers(min_value=0,
                   max_value=10_000_000))
```

  change the implementation using an exact algorithm `https://en.wikipedia.org/wiki/Integer_square_root`

Properties in Hypothesis

- Properties are functions that fail if some expected condition is not met
- Arguments are universally quantified
- For each property:
  - will be test with a large number of random values (100 by default)
  - generated using strategies (defined by the `@given` decorator)
- Module `hypothesis.strategies` provides:
  - predefined strategies for basic types
  - methods for modifying and combining strategies

Strategies for basic values

floats()	generate floating-point numbers
integers()	generate integers
booleans()	generate logical values
text()	generate Unicode strings

You can:

- customize some strategies using parameters (e.g. `min_value` and `max_value` for numbers)
- inspect examples of values drawn from a strategy using the `example` method

Strategies for basic values (cont.)

```
>>> from hypothesis.strategies import integers, booleans
>>> integers(min_value=0, max_value=10).example()
4
>>> integers(min_value=0, max_value=10).example()
7
>>> booleans().example()
True
>>> booleans().example()
False
```

Mapping and filtering strategies

We can use the `map` and `filter` methods to adapt strategies:

- **map** applies a function to every value generated
- **filter** discards values that do not meet a specified condition

Mapping and filtering strategies (cont.)

Examples:

```
# generate only even numbers
integers().map(lambda x:2*x)

# generate only odd numbers
integers().map(lambda x:2*x+1)

# generate only even numbers, alternative
integers().filter(lambda x:x%2==0)
```

Mapping and filtering strategies (cont.)

Caveat:

- `filter` may need to discard many generated values
- if the condition is rarely met, filtering will be slow
- Hypothesis will give up if filtering takes too long

```
>>> integers(max_value=10).filter(lambda x: x>10).example()
hypothesis.errors.Unsatisfiable: ...
```

Strategies are first-class values

- We can give names to strategies and use them later
- This allows defining strategies compositionally and improves test readability

Strategies are first-class values (cont.)

```
from hypothesis import given
import hypothesis.strategies as st

# a strategy for odd integer numbers
odds = st.integers().map(lambda x:2*x+1)

# a simple property:
# adding two odd numbers gives an even number
@given(odds, odds)
def test_add_odd_numbers(x, y):
    assert (x+y)%2 == 0  # check that the sum is even
```

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Integrating Hypothesis with Python tools

- Python `pytest` tool can pick up Hypothesis tests as well as ordinary unit tests
- We can use VSCode `pytest` integration to automatically run property tests
- Tips:
  - Open the source directory with Python files
  - Define tests in files with `test_` prefix
  - Define properties with `test_` prefix

(Cue demo.)

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Practice session 1

1 Test the integer square test property and fix the failure by limiting the integers generated

2 Introduce an error into the implementation that gives the wrong answer for some arguments e.g.:

```
def isqrt(n):
    if n>1000 and n<2000:
        return n//2   # WRONG
    else:
        return int(math.sqrt(n))
```

Re-run the Hypothesis tests and see what counter-example you get.

Practice session 1 (cont.)

Define and test the following properties using Hypothesis.

3 Adding two even integer numbers always gives an integer result.

4 Adding an even integer number and an odd integer number gives an odd integer number.

5 Integer addition is *commutative*: $x + y$ should always give the same result as $y + x$.

6 Integer addition is *associative*: $(x + y) + z$ should always give the same result as $x + (y + z)$.

7 Floating point addition is associative (spoiler: it is not and Hypothesis will produce a counter-example).

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Strategies for compound data

lists(s) lists of elements given by strategy s

tuples($s1, s2, \ldots$) lists of elements given by strategies $s1, s2, \ldots$

dictionaries($s1$, $s2$) dictionaries with keys given by strategies $s1$ and values given by $s2$

These strategies take optional parameters to control generation (see the documentation).

Strategies for compound data (cont.)

Some examples:

```
# lists of integers from 0 to 100
lists(integers(min_value=0,max_value=100))

# lists of integers with at least 3 elements
lists(integers(), min_size=3)

# tuples of a boolean and an integer
tuples(booleans(),integers())

# dictionaries from 5 character strings to integers
dictionaries(texts(min_size=4, max_size=4), integers())
```

A simple property on lists

Reversing a list twice should give back the original list.

```
from hypothesis import given
import hypothesis.strategies as st

@given(st.lists(st.integers()))
def test_reverse_twice(x):
    y1 = list(reversed(x))   # first reverse
    y2 = list(reversed(y1)) # second reverse
    assert y2 == x
```

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Shrinking

- Hypothesis attempts to simplify the counter-examples found before presenting
- This is useful to remove "noise" from randomly generated data
- Each strategy employs a specific criteria for simplification
- For example, `lists()` attempts to:
  - reduce the list length
  - if that is not possible, shrink some values inside the list
- Hypothesis keeps shrinking while the example still causes the property to fail

An example

A wrong property on lists:

```
@given(st.lists(st.integers()))
def test_reverse_wrong(x):
    assert list(reversed(x)) == x
```

An example

A wrong property on lists:

```
@given(st.lists(st.integers()))
def test_reverse_wrong(x):
    assert list(reversed(x)) == x
```

```
pytest basic.py -k test_reverse_wrong
```

```
E        Falsifying example: test_reverse_wrong(
E            x=[0, 1],
E        )
```

An example

A wrong property on lists:

```
@given(st.lists(st.integers()))
def test_reverse_wrong(x):
    assert list(reversed(x)) == x
```

```
pytest basic.py -k test_reverse_wrong
```

```
E        Falsifying example: test_reverse_wrong(
E            x=[0, 1],
E        )
```

- The "smaller" lists `[]`, `[0]`, `[1]` and `[0,0]` are *not* counter-examples
- The counter-example reported is *minimal*
- In general, shrinking will find local minima, not the global minimum

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Composing strategies

- We can use compound strategies to build new strategies for domain-specific applications
- For simplicity, we will only look at the basics in this session
- Hypothesis provides more advanced combinators for building strategies (check my book for more advanced uses)

Composing strategies (cont.)

Example: a strategy for generating dates.

- years from 1970 to 2100
- months from 1 to 12
- days from 1 to 31

Composing strategies (cont.)

First attempt:

```
years = integers(min_value=1970, max_value=2100)
months = integers(min_value=1, max_value=12)
days = integers(min_value=1, max_value=31)
dates = tuples(years, months, days)
```

Problem: will generate invalid dates e.g. 30th of February.

Composing strategies (cont.)

Solution: filter invalid dates.

```
years = integers(min_value=1970, max_value=2100)
months = integers(min_value=1, max_value=12)
days = integers(min_value=1, max_value=31)
dates = tuples(years, months, days).filter(valid_date)

def valid_date(date):
    (y,m,d) = date
    # check if the date is valid
    # and return True or False
    ...
```

Note: in real code you should use the `dates` strategy from the Hypothesis library.

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A larger example

Encode and decode to *comma-separated values* (CSV) a table with employee data:

- a numerical identifier
- the employee's name
- the starting work date
- a text field for notes

A larger example (cont.)

Example:

ID	Name	Start Date	Notes
1	Alice Johnson	2022-01-15	Enjoys Python
2	Bob Lee	2021-11-02	Fluent in French
3	Clara Smith	2020-05-30	Handles onboarding, "HR"
4	Dan O'Neil	2023-03-18	Loves branding, PR

A larger example (cont.)

- We will represent the table in Python as a list of tuples
- The `encode` and `decode` functions convert between the Python representation and a CSV string
- Implemented using Python's `csv` library (check the code repository)

```
>>>> encode([(1,'Alice',date(2022,1,15),'Enjoys Python')])
'1,Alice,2022-01-15,Enjoys Python\r\n'
>>>> decode('1,Alice,2022-01-15,Enjoys Python\r\n')
[(1, 'Alice', datetime.date(2022, 1, 15), 'Enjoys Python')]
```

A larger example (cont.)

A round-trip property: encoding followed by decoding should preserve information.

```
ids = st.integers(min_value=1, max_value=100)
names = st.text(min_size=1)  # avoid empty names
starts = st.dates(min_value=date(2000,1,1),
                  max_value=date(2025,1,1))
notes = st.text()
rows = st.tuples(ids, names, starts, notes)
tables = st.lists(rows, min_size=1)  # avoid empty tables

@given(tables)
def test_csv_round_trip(table):
    assert decode(encode(table)) == table
```

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Practice session 2

8 Run the test dates example from the code repository to confirm that our `dates` can generate invalid dates.
Observe how Hypothesis uses shrinking to find the simplest counter-example.
Remove the shrinking phase from the test settings and re-run the tests.

9 Use a roundtrip property of the CSV encoding example to test the "naive" encode and decode functions given in the repository:

```
@given(tables)
def test_csv_naive_round_trip(table):
    assert decode_naive(encode_naive(table)) == table
```

The tests for the naive functions will fail; can you interpret the counter-examples found?

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Testing stateful programs

We can also use Hypothesis to test code that maintain state, e.g:

- stateful objects;
- web services;
- database connections.

Rule-based State Machine Testing

- Start from a well-defined initial state
- Each transition (called a rule) can take arguments generated by strategies
- Transitions are executed both in the implementation and the model
- Compare the results of implementation against the model after each step

$$
\begin{tikzcd}
\text{implementation} & s_0 \arrow[r, "r_1"] & s_1 \arrow[r, "r_2"] \arrow[d, dash, "?"] & s_2 \arrow[r, "r_3"] \arrow[d, dash, "?"] & s_3 \arrow[r, "r_4"] \arrow[d, dash, "?"] & \cdots \\
\text{model} & m_0 \arrow[r, "r_1"] & m_1 \arrow[r, "r_2"] & m_2 \arrow[r, "r_3"] & m_3 \arrow[r, "r_4"] & \cdots
\end{tikzcd}
$$

Rule-based State Machine Testing (cont.)

- Tests fail if there is a mismatch between model and implementation
- The counter example is the rule sequence leading to the failure
- Shrinking is applied to find a minimal sequence

$$
\begin{tikzcd}
\text{implementation} & s_0 \arrow[r, "r_1"] & s_1 \arrow[r, "r_2"] \arrow[d, dash, "\checkmark"] & s_2 \arrow[r, "r_3"] \arrow[d, dash, "\checkmark"] & s_3 \arrow[d, dash, "\times"] \\
\text{model} & m_0 \arrow[r, "r_1"] & m_1 \arrow[r, "r_2"] & m_2 \arrow[r, "r_3"] & m_3
\end{tikzcd}
$$

Sequence $[r1, r2, r3]$ leads to a test failure.

First example

- A database for associating *users* and *emails*
- Implemented as a Python class and a SQLite backend
- Users are represented by an integer id
- Each user can have more than one email
- Operations:

  add(userid,email) add an email to a user
  get(userid) get all emails from a user
  remove(userid,email) remove and email from a user
  delete(userid) remove all emails from a user

First example (cont.)

```
>>> db = UserDB()
>>> db.add(1, 'foo@bar.com')
>>> db.add(2, 'baz@bong.org')
>>> db.get(1)
['foo@bar.com']
>>> db.add(1, 'foo@bing.org')
>>> db.add(1, 'foo@bar.com')
>>> db.get(1)
['foo@bar.com', 'foo@bing.org']
>>> db.remove(1, 'foo@bar.com')
>>> db.get(1)
['foo@bing.org']
>>> db.close()
```

Testing the User Database

- The model is a list of pairs (user,email)
- The state machine starts with an empty DB and empty model
- Each transition is defined as a method labeled `@rule`
  - corresponds to one of the UserDB operations
  - performs the operation in both implementation and model
  - checks that the implementation and model agree
- Hypothesis will generate random sequences of rules
- A counter-example will be a sequence of rules leading to a mismatch

(Cue demo.)

Testing the User Database (cont.)

```
AssertionError:
assert ['0@B.ac', '0@A.ac'] == ['0@A.ac', '0@B.ac']
Falsifying example:
        state = TestUserDB()
        state.add(email='0@B.ac', user=1)
        state.add(email='0@A.ac', user=1)
        state.get(user=1)
        state.teardown()
```

Fixing the testing failure

- Hypothesis found a mismatch between the implementation and model: the order of emails obtained is not always in insertion order
- We can correct this in two ways:

  strengthening the implementation fixing the order results in the SQL query

  weakening the property allow results with different orders
- More generally:
  - testing failures show that an assumption about our code does not hold
  - the error could be in the assumption, the code or both

Second example

- A simplified shopping cart for an on-line store
- Maintains quantities associated with each item
- Operations:

init(prices)	define items and prices
add(item)	add an extra item
remove(item)	remove an item (if possible)
get_subtotal()	compute current subtotal i.e. sum prices times quanties

Second example (cont.)

```
>>> c = ShoppingCart({'apples':10,'pears':20,'berries':30})
>>> c.add('apples')
>>> c.add('bananas')
ValueError: invalid item
>>> c.add('berries')
>>> c.add('apples')
>>> c.get_subtotal()
50
>>> c.remove('berries')
>>> c.get_subtotal()
20
```

Testing the shopping cart

- Unlike the previous example, we will not define an explicit model
- Instead we will just state post-conditions of each of the operations:

add	the sub-total should increase by the item's price
remove	the sub-total should decrease by the item's price

(Cue demo.)

Understanding the testing failure

```
AssertionError: assert 0 == (0 - 1)
Falsifying example:
    state = ShoppingCartMachine()
    state.start(price_dict={'A': 1, 'B': 1, 'C': 1,
                            'D': 1, 'E': 1, 'F': 1})
    state.add(item='E')
    state.remove(item='E')
    state.remove(item='E')
    state.teardown()
```

- Testing found a sequence that violated our post-conditions
- The error is caused by a bug in the implementation of the `remove` method

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10 Correct the error in the user email database by changing the post-condition of the `get` rule to sort the results before comparing:

```
assert sorted(answer) == sorted(expect)
```

Confirm that the tests pass with this modification.

11 Correct the error in the `remove` method of the shopping cart class so that it deletes an entry when the quantity drops to zero.

Practice session 3 (cont.)

12 Consider two extra methods for the shopping cart class in the code repository:

save exports the contents of the shopping cart to a string;

load imports the contents of the shopping cart from a string.

Modify the state machine to test these methods using the following idea:

1. define two additional fields, namely, `saved_cart` and `saved_subtotal` initially both set to `None`;
2. add a rule for save that sets the `saved_cart` and `saved_subtotal`;
3. add a rule for load with a precondition that `saved_cart` is not `None`; this should load the saved cart and verify that the subtotal equals the `saved_subtotal`.

This allows for tests that introduce some arbitrary number of cart operations between loads and saves. Check that the tests still pass.

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Conclusion

- PBT philosophy: write properties and generate tests
- Lightweight: implemented as a library
- Flexible: a "domain-specific language" for writing strategies and properties
- Properties become "executable specifications" of the code
- Useful for finding subtle bugs in complex systems
- Useful for comunicating expectations among developers
- Conjecture: is validation of AI generated code a "killer app" for PBT?

More information

- Hypothesis documentation. `https://hypothesis.readthedocs.io/`
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